

MONTE REAL AB, Portugal

WMO: 085400

Lat: 39.828N

Lon: 8.883W

Elev: 54

StdP: 100.68

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	2.0	-2.2	3.2	4.8	-0.8	3.6	5.1	10.4	11.4	9.2	11.8	1.3	160	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	31.3	19.6	28.9	19.0	26.6	18.5	21.2	27.0	20.4	25.5	19.8	24.2	4.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	22.5	19.0	13.9	21.8	18.2	13.2	21.3	61.4	26.9	58.6	25.3	56.7	24.4	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.3	-1.8	35.7	1.2	2.2	-2.7	37.3	-3.4	38.6	-4.1	39.8	-5.0	41.4		
			-2.2	22.9	1.4	1.6	-3.1	24.1	-3.9	25.0	-4.7	25.9	-5.7	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	10.4	10.8	12.7	14.4	16.4	18.5	20.0	20.4	19.5	17.6	12.8	11.0
	DBStd	4.42	2.57	2.55	2.54	2.63	2.69	2.27	1.99	2.04	2.16	2.78	3.04	2.83
	HDD10.0	84	27	19	6	1	0	0	0	0	0	0	9	23
	HDD18.3	1303	246	210	174	120	74	23	3	2	12	47	166	228
	CDD10.0	2058	39	42	91	133	198	256	309	322	285	235	94	53
	CDD18.3	238	0	0	1	3	14	28	54	66	47	23	1	0
	CDH23.3	1492	0	0	16	48	143	194	278	393	263	153	4	0
	CDH26.7	494	0	0	2	8	43	66	113	146	78	38	0	0
Wind	WSAvg	3.2	2.9	3.2	3.3	3.4	3.5	3.5	3.8	3.5	2.8	2.8	2.8	3.0
Precipitation	PrecAvg	781	114	81	64	70	62	19	8	13	40	110	113	100
	PrecMax	1131	298	196	193	218	141	62	35	43	112	234	390	318
	PrecMin	428	7	10	7	29	1	1	0	0	4	10	12	4
	PrecStd	179	74	54	52	43	39	18	9	12	32	56	81	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	20.8	25.2	28.0	31.0	32.2	34.5	34.7	32.0	30.3	23.5	19.1
		MCWB	13.9	13.5	14.6	18.0	19.0	20.0	21.3	21.9	19.2	18.0	16.6	13.9
	2%	DB	16.6	18.5	22.3	24.3	27.2	28.2	29.9	30.8	28.9	27.0	20.9	17.6
		MCWB	13.4	12.7	13.9	15.8	17.9	18.9	19.5	20.1	19.2	18.0	15.7	14.2
	5%	DB	15.7	16.8	19.9	21.9	24.1	25.1	25.7	27.2	26.1	24.8	19.0	16.5
		MCWB	13.3	12.5	13.4	15.3	17.0	18.5	19.4	19.4	18.8	17.4	15.7	13.7
	10%	DB	14.8	15.3	17.3	19.2	21.3	22.9	23.8	24.9	24.1	22.5	17.8	15.6
		MCWB	12.9	12.5	13.1	14.7	15.9	17.7	18.7	19.2	18.5	17.3	15.0	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.2	16.9	18.7	20.2	21.1	22.7	23.0	21.2	21.0	19.1	16.2
		MCDB	16.6	16.9	21.7	25.0	27.8	28.4	31.6	31.0	26.5	25.4	20.4	17.3
	2%	WB	14.7	14.4	15.5	17.0	18.8	20.0	20.9	21.2	20.3	20.0	17.4	15.4
		MCDB	15.7	16.3	19.2	22.3	24.9	25.8	26.4	26.9	25.3	23.3	19.2	16.5
	5%	WB	14.0	13.8	14.5	15.8	17.5	19.0	19.9	20.4	19.6	19.2	16.4	14.5
		MCDB	15.1	15.8	17.9	20.3	22.4	24.4	24.2	25.1	24.1	22.2	18.2	15.9
	10%	WB	13.3	12.9	13.7	15.0	16.5	18.2	19.2	19.6	19.0	18.3	15.4	13.6
		MCDB	14.4	15.1	16.7	18.5	20.3	22.1	22.8	23.7	22.9	21.4	17.3	15.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	8.8	8.6	8.9	8.4	8.4	9.5	9.8	9.8	8.8	8.5
		MCDBR	9.2	11.3	14.2	13.8	14.4	13.7	13.6	15.2	14.1	14.0	10.5	8.7
		MCWBR	6.3	6.7	7.2	6.8	6.6	6.0	5.7	6.4	6.0	6.4	6.3	5.9
	5% WB	MCDBR	5.9	7.8	10.6	11.3	11.9	11.2	10.6	11.3	10.4	9.7	7.0	6.1
		MCWBR	4.9	5.5	6.4	6.3	6.2	5.5	5.1	5.4	5.1	5.4	4.9	5.1
Clear-Sky Solar Irradiance	taub	0.332	0.345	0.377	0.375	0.382	0.387	0.375	0.388	0.383	0.369	0.342	0.330	
	taud	2.496	2.453	2.368	2.386	2.394	2.399	2.423	2.391	2.424	2.472	2.503	2.500	
	Ebn at Noon	833	872	876	900	896	889	897	876	859	833	814	804	
	Edh at Noon	78	92	111	116	118	118	114	114	104	90	77	72	
All-Sky Solar Radiation	RadAvg	2.07	3.07	4.16	5.40	6.51	7.07	7.24	6.58	5.12	3.44	2.31	1.87	
	RadStd	0.30	0.39	0.51	0.55	0.55	0.41	0.33	0.28	0.37	0.32	0.34	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page